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Introduction to
*Asset Pricing and Portfolio Performance:
Models, Strategy, and Performance Metrics*

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Introduction and Overview

A well-established empirical observation is that different classes of investment vehicles have earned very different average returns. For example, over the seventy-three year period from the end of 1925 to the end of 1998 large-capitalization stocks (those in the Standard & Poors index) earned an average annual total return of 13.2%. Over the same period, a portfolio of small-capitalization stocks earned 17.4% per year, a portfolio of long-term corporate bonds earned 6.1% per year, a portfolio of long-term government bonds earned 5.7% per year, a portfolio of intermediate-term government bonds earned 5.5% per year, and short-term government bonds earned 3.8% per year [Ibbotson Associates (1999, Table 2-1)]. Similar comparisons could be made across different national markets, among different individual assets within an asset class (e.g., IBM versus GE common stock), across different managed portfolios, such as mutual funds, or across different simulated portfolio strategies.

A critical question is: “How should one interpret such differences in returns?” Are stocks better investments than bonds since they have higher returns? Is mutual fund A better than mutual fund B merely because it historically has higher average returns? How should we choose between disparate portfolio strategies such as small-capitalization stocks versus large-capitalization stocks, value stocks versus growth stocks, or momentum vs. contrarian policies? The papers included here develop a theoretical framework for evaluating observed differences in returns across assets and present empirical evidence on the models’ ability to fit observed asset returns. The main themes that the works address are the appropriate adjustment for risk differences across assets, the pricing of market friction-induced characteristics (such as liquidity), and inference in the presence of randomness.

Certainly some of the differences in returns across assets, asset classes, and portfolios may be due to randomness, or luck. However, luck cannot explain all these observed differences. For example, standard statistical methods indicate that many of the differences in average returns, across the asset classes cited above, are unlikely to occur by pure chance.

One clear difference across assets is the level of risk. One measure of risk is the volatility in returns, as measured by the standard deviation, or equivalently the variance, of the return distribution. For the asset classes and time period cited above, the sample standard deviation of annual returns is 20.3% for large-capitalization stocks, 33.8% for small-capitalization stocks, 8.6% for corporate bonds, 9.2% for long-term government bonds, and 5.7% for intermediate-term government bonds [Ibbotson Associates (1999, Table 2-1)]. While the common stock portfolios are clearly more volatile and earn a higher returns than the bonds, there is not a simple relation between average returns and sample standard deviations. The primary role of asset pricing models has been to specify the appropriate measure(s) of risk and the appropriate risk/return tradeoff.

Another important difference among assets that may influence anticipated returns is the effects of market frictions on the pricing of assets. For example, the liquidity of an asset, or the extent to which it can be purchased or sold without a large premium or discount, may influence its market clearing price. For example, Daves and Ehrhardt (1993) find that U.S. Treasury bonds with identical cash flows sell at different prices. They argue that the difference in pricing is due to differences in liquidity.

The literature on asset pricing models and portfolio performance metrics is specifically aimed at disentangling the effects of luck, skill, risk, and liquidity. The models provide a

benchmark return against which the returns on individual assets, and passive, active, or simulated portfolios can be judged. Much of the theory and empirical testing has focused on the appropriate way to measure risk, and discerning when return differences could easily be due to luck. More recently, the role of non-risk characteristics, such as liquidity, has received increased attention.

While the above evidence looks at long-run, or unconditional, differences in asset returns, there is no reason why expected returns across assets need be constant through time. Even if risk premia are constant through time, the expected return on an active or dynamically rebalanced portfolio could change through time as the risk of the portfolio changes. Such dynamic rebalancing could be due, among other things, to trading on private information, to non-information-based rebalancing [e.g., Constantinides (1986)], or to dynamic trading designed to replicate derivative securities. Unconditional asset pricing models are not well-suited for evaluating some types of portfolio trading strategies and need to be modified to reflect the time-varying nature of risks or risk premia.

The volume begins with a section entitled **Asset Pricing Theory** consisting of papers that established the foundation of asset pricing theory in the form of the Capital Asset Pricing Model (CAPM), the Intertemporal Capital Asset Pricing Model (ICAPM), and the Arbitrage Pricing Theory (APT). Additionally, a version of the CAPM that incorporates incomplete information is included.

The papers in the next section, **Testing Asset Pricing Models, Anomalies, and Portfolio Strategies**, subject the theories of the preceding section to empirical tests. The testable implications of the frictionless market-based models are that only certain measures of

risk command a risk premium. Years of empirical testing have led to a number of apparent deviations between the predictions of the asset pricing models and actual returns. These anomalous results suggest potentially profitable portfolio trading strategies. That is, if one can predict returns based on variables not associated with risk, then one can form portfolios that have higher return without higher risk, thus beating the benchmark provided by the asset pricing model.

The papers in **Market Imperfections and Asset Pricing** look at the effects of market frictions such as illiquidity, transactions costs or non-insurability of some risks on the pricing of securities.

The papers in the section titled **Portfolio Performance Evaluation** use asset pricing models to disentangle the elements of risk, skill, and luck in the evaluation of portfolio performance. In particular, the papers chosen stress the effects of active management and dynamic portfolio trading strategies on the problem of inferring the skill of the portfolio manager.

The body of work in this volume is designed to answer the question with which we started : “How should one interpret differences in returns?” First, the papers develop a set of models which tell us how to measure risk and how returns should be related to that risk. Next, we move on to testing the models’ predictions and addressing the role of market frictions in asset pricing. The tests of the models’ predictions and portfolio performance metrics derived from the asset pricing models are designed to distinguish between random performance (luck), performance due to risk taking (risk), and superior portfolio construction (skill).

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Asset Pricing Theory

The foundation of modern asset pricing theory was Harry Markowitz's theory of portfolio selection [Markowitz (1952, 1959, 1987)]. Markowitz (1952) characterizes the portfolio selection problem for investors choosing mean-variance efficient portfolios (portfolios that have the highest expected rate of return for a given level of volatility, as measured by variance or standard deviation).¹ A clear result from this analysis is that, even in the case where investors measure the risk of their portfolio by its variance, variance cannot be the determinant of the expected returns on individual assets. A simple counter example can illustrate this fact. Suppose that an asset's variance determines its expected returns. Consider two assets that have the same variance. If variance determines expected returns, these two assets will have the same expected return. If we construct a portfolio that has a 50% investment in each of the two assets, the effects of diversification imply that the variance on this portfolio will be less than the variance of the two component assets (unless the returns on the two assets are perfectly correlated). However, the expected return on the portfolio must, by definition, equal a weighted average of the returns on the component assets. This contradicts the original assumption about the relation between variance and expected returns.

Equilibrium expected rates of return need to be considered in the context of the investor's optimal portfolio selection problem. Assuming a frictionless, single-period world in which

¹Other objective functions are considered in Markowitz (1959).

investors rank portfolios in terms of their common beliefs about the first two moments (mean and variance, or standard deviation) of the portfolios' probability distributions, investors optimally hold mean-variance efficient portfolios. When a riskless asset is available to investors, all of the efficient portfolios are perfectly correlated with each other.

The papers by **Sharpe (1964)**, **Treynor (1961)**, Lintner (1965), and Mossin (1966) study the implications of competitive equilibrium in this securities market for the measurement and pricing of risk. Their analysis shows that the total risk (variance) of an asset can be decomposed into a component that is correlated with the mean-variance efficient portfolios, and a component that is uncorrelated with those portfolios. They find that the former component commands a risk premium while the latter component does not. **Sharpe (1964)** refers to the two components as systematic and non-systematic risk, while **Treynor (1961)** refers to them as uninsurable and insurable risks. Also, in equilibrium, the market portfolio of all available assets must be one of these efficient portfolios. The appropriate measure of priced risk for asset i is its "beta" with respect to the aggregate market portfolio, denoted $\beta_{i,m}$. Beta is equal to the correlation coefficient between the return on asset i and the return on the market portfolio, scaled by the ratio of the standard deviation of the return on asset i and the standard deviation of the return on the market portfolio. It is also the slope coefficient of a regression of asset i 's return on the return of the market portfolio.

The relation between expected returns and beta, the **Capital Asset Pricing Model (CAPM)**, derived in these papers, is:

$$E(r_i) = r_f + \beta_{i,m}[E(r_m) - r_f]$$

where $E(\bullet)$ denotes expectation, r_i is the return on asset i , r_f is the riskless interest rate, r_m is the return on the market portfolio. The expected return is given by the pure time-value of money, r_f , plus premium for bearing risk, $\beta_{i,m}[E(r_m) - r_f]$. The premium is given by the level of risk, $\beta_{i,m}$, times the market risk premium, $[E(r_m) - r_f]$. Investors do not care about an asset's variability that is uncorrelated with their optimal portfolio since they do not bear any of that risk.

The CAPM gives a simple benchmark against which we can compare the returns on assets, asset classes, and portfolios. Assets or portfolios with larger a beta should earn a higher level of return in order to compensate for the added risk.

The fundamental contributions to the CAPM of **Sharpe (1964)** and **Treynor (1961)** are included here. It is a pleasure to be able to publish, for the first time, Jack Treynor's paper, which has been circulating in manuscript form for nearly forty years.

Like all economic models, the CAPM was derived under a set of simplifying assumptions. Over the years there have been a number of extensions to the CAPM, prompted both by the desire to have a model based on less restrictive assumptions and by empirical evidence indicating deviations from the CAPM predictions.

Early extensions of the CAPM include versions incorporating the effects of taxation [Brennan (1970)], eliminating the assumption of a riskless asset [Black(1972)], and incorporating the pricing of higher-order moments of return distributions [Ingersoll (1975) and Kraus and Litzenberger (1976)]. The single period nature of the original model can be justified if agents' future consumption and investment opportunities are non-random and their tastes for goods and services are state independent [Fama (1970)]. The third paper in this volume, **Merton (1973)**, develops the Intertemporal Capital Asset Pricing Model (ICAPM) in which investors can

trade assets continuously through time and in which future investment opportunity can change through time. The equilibrium risk premia on assets are determined by their correlation with the market portfolio *and* by their correlation with portfolios that mimic shifts in the consumption/investment opportunity sets. If we assume there are k sources of uncertainty in the consumption/investment opportunity set, Merton's analysis leads to a multiple-beta asset pricing model of the form:

$$E(r_i) = r_f + \beta_{i,m}[E(r_m) - r_f] + \beta_{i,1}[E(r_{h1}) - r_f] + \beta_{i,2}[E(r_{h2}) - r_f] + \dots + \beta_{i,k}[E(r_{hk}) - r_f]$$

where $r_{h1}, r_{h2}, \dots, r_{hk}$ are the returns on k portfolios that are maximally correlated with the k sources of uncertainty in the consumption and investment opportunity sets (hedge portfolios) and the beta coefficients, $\beta_{i,m}, \beta_{i,1}, \beta_{i,2}, \dots, \beta_{i,k}$ are multiple regression coefficients that measure the sensitivity of asset i to the $k + 1$ sources of risk [also see Roll (1973) and Long (1974)]. In general, the expected returns and risk parameters, $\beta_{i,m}, \beta_{i,1}, \beta_{i,2}, \dots, \beta_{i,k}$, will vary through time as the conditional probability distribution of asset returns changes. If the conditional distribution is constant the betas in the pricing model will be constant. **Merton** provides an example in which the risk of shifts in the riskless rate of interest is the only source of risk in the investment opportunity set ($k = 1$).

The Arbitrage Pricing Theory (APT), derived in the paper by **Ross**, provides an alternative approach to obtaining a multiple-beta asset pricing model, at least as an approximation. **Ross** assumes there is a infinite number of assets and that returns on the assets follow a factor model:

$$r_i = E(r_i) + \beta_{i,1}\delta_1 + \beta_{i,2}\delta_2 + \dots + \beta_{i,k}\delta_k + \varepsilon_i$$

where $E(\delta_j) = E(\varepsilon_i) = 0$ for $j = 1, \dots, k$ and all i , $E(\varepsilon_i \varepsilon_p) = 0$ for $i \neq p$, and the variance of ε_i is bounded for all i . The interpretation of the factor model is that asset returns are driven by k common sources of uncertainty (the deltas) as well as asset specific sources of uncertainty (the epsilons).

An asymptotic arbitrage opportunity is the ability to construct a sequence of zero-investment, zero-beta portfolios whose risk approaches zero, but whose returns approach infinity. Such opportunities should be competed away by traders. The lack of asymptotic arbitrage opportunities places a restriction on the expected returns of the assets:

$$E(r_i) \approx r_f + \beta_{i,1}[E(r_{\delta 1}) - r_f] + \beta_{i,2}[E(r_{\delta 2}) - r_f] + \dots + \beta_{i,k}[E(r_{\delta k}) - r_f]$$

where $r_{\delta 1}, r_{\delta 2}, \dots, r_{\delta k}$ are the returns on k portfolios that are maximally correlated with the k factors, $\delta_1, \delta_2, \dots, \delta_k$. Let $\alpha_i = E(r_i) - \{r_f + \beta_{i,1}[E(r_{\delta 1}) - r_f] + \beta_{i,2}[E(r_{\delta 2}) - r_f] + \dots + \beta_{i,k}[E(r_{\delta k}) - r_f]\}$, the deviation of asset i 's expected return from the linear asset pricing model. The approximation comes from the fact that the APT puts a bound on the sum of the squared deviations:

$$\sum_{i=1}^{\infty} \alpha_i^2 < \infty$$

While this implies that most assets' expected returns will be very close to the linear asset pricing model, there could be a finite set of assets with large deviations from the model. With some ancillary assumptions outside of the original APT formulation [e.g., Connor (1984)] the pricing model will hold as an equality. The APT combined with these ancillary assumptions is generally referred to as equilibrium versions of the APT. In practice, most applications ignore the approximation and treat the APT pricing relation as exact rather than an approximation. As such, they can be viewed as applying the equilibrium APT. Like the original CAPM, the APT is a single period asset pricing model. It can be extended to an intertemporal setting [Chamberlain (1988), Connor and Korajczyk (1989), and Constantinides (1989)] and to accommodate conditioning information [Stambaugh (1983)].

It is clear that the form of the ICAPM and APT pricing relations is the same. In the ICAPM, the priced risk factors are state variables that are correlated with future shifts in consumption and investment opportunities in addition to the aggregate market portfolio. In the APT the priced risk factors are sources of common, pervasive shocks to asset prices. In practice, the distinction between the two models has become somewhat blurred. For example, empirical multi-factor models that are motivated by appeals to ICAPM hedging demands are often identified as APT-based models.

In addition to the models discussed above, there is a large body of literature on consumption-based asset pricing models (CCAPM) [Lucas (1978), Breeden (1979), Brock (1982)]. This literature is much more important than the space allocated to it in this volume might indicate. For what are primarily data issues, the portfolio performance literature has not

tended to rely extensively on the CCAPM literature. Given that the thrust of this volume is asset pricing and portfolio performance, the choice of coverage reflects this tendency.

The equilibrium models discussed so far have relied on the assumption that access to information is homogeneous across investors. The second paper by **Merton** relaxes this assumption by assuming that each investor is informed about the distribution of returns on a subset of assets in the economy. **Merton (1987)** also assumes that investors only invest in assets about which they are informed [also see Levy (1978)].

With symmetric information, **Merton's** model collapses to the CAPM. With asymmetric information there is an additional premium or discount that reflects the extent to which investors are informed about that asset versus all assets in the market. Other things being equal, assets with a greater base of informed investors will have higher prices, and therefore, lower expected returns.

Testing Asset Pricing Models, Anomalies, and Portfolio Strategies

The asset pricing models in Section I provide predictions of the appropriate measure of risk and the form of the risk/return relation. Each model gives us a benchmark against which we can compare realized returns on assets. In the introduction to Section I, α_i was defined to be the difference between asset i 's expected return and the return predicted by the asset pricing model. The CAPM, ICAPM, and equilibrium versions of the APT each imply that α_i should be zero. Thus, α_i represents an abnormal return in those models. **Merton's** incomplete information CAPM implies that the α_i from the CAPM should be related to a number of factors other than

systematic risk, such as the fraction of investors informed about the stock and its unsystematic risk.

The papers in this section will subject the models to empirical testing by asking whether α_i is forecastable either across assets or across time for a given asset.. The literature finds that there are seemingly persistent predictable deviations between the returns on assets and the predictions of the models. That is, the estimates of the pricing deviation variable, $\hat{\alpha}_i$, seem to be forecastable. If true, these pricing anomalies imply that one could construct portfolios that beat the benchmark [Treyner and Black (1973)].

The literature on testing asset pricing models is huge. Rather than choose a set of papers that reflects each of the paths the literature has taken (i.e., often one anomaly and one asset pricing model at a time), I have chosen to include primarily papers that simultaneously look at a collection of anomalies and/or compare the performance of multiple models. An excellent survey that provides a more complete chronology of the literature on asset pricing anomalies is Hawawini and Keim (1995).

In this section's first paper, **Fama and French (1992)** study the performance of the CAPM against a number of firm characteristics that had been identified in the literature as having predictive ability for returns (firm size, leverage, book-to-market equity ratio, earnings to price ratio). The CAPM predicts that systematic risk, $\beta_{i,m}$, should explain cross-sectional differences in returns and that other, non-risk-based, predetermined variables should not. **Fama and French** find that $\beta_{i,m}$ has no significant explanatory power while a number of the predetermined variables do. In particular, two variables, firm size (as measured by the market

value of equity) and the ratio of the book value of equity to the market value of equity, have strong explanatory power.

These results indicate that cross-asset differences in the deviation from CAPM pricing, α_i , can be reliably predicted. Additionally, the nature of the deviations from the CAPM can be identified with particular types of portfolio strategies. Smaller firms have positive values of α_i while larger firms have negative values. This suggests a strategy of forming portfolios tilted toward small-capitalization firms. Likewise, high (low) book-to-market firms tend to have positive (negative) values of α_i . This suggests a strategy of forming portfolios tilted toward value firms, as opposed to growth firms.

Another common characterization of alternative portfolio strategies is contrarian versus relative strength (momentum). A contrarian strategy is one that purchases assets that have lost value and sells assets that have gained value. Conversely, a momentum strategy is one that purchases assets that have gained value and sells assets that have lost value. The paper by **Jegadeesh and Titman** tests the ability of contrarian and momentum strategies to generate non-zero values of α_i . They find that momentum strategies lead to significantly positive values of α_i for a period of one year after forming the portfolio. The pricing deviation turns negative in the second year after portfolio formation and is zero thereafter.

Thus, there is a fair amount of evidence that indicates significant deviations from the CAPM predictions. In particular the success of size, book-to-market, and momentum strategies has been difficult to explain. When faced with evidence against a particular model, there are several possible explanations:

- a. The model is wrong and we need to use a better model.

- b. The model is fine but we have applied it incorrectly. For example, in testing the CAPM we may have not use the correct market portfolio [Roll (1977)]. Also, the risk parameters may be changing through time such that the prediction variables are correlated with the estimation error in beta.
- c. The anomalous evidence is an artifact of faulty data or statistical techniques. For example, the data might be tainted by selection or survival biases, making portfolio strategies seem more profitable than they are in reality. A related issue is one of data mining. If we test many portfolio strategies on the same data set, some will look statistically significant merely by chance.

The remaining papers in this section and the papers in the next section address these issues.

A logical direction to take is to consider the ability of the multi-factor models discussed in Section I, the ICAPM and APT, to explain the deviations from the CAPM. A critical issue with these models is the choice of the set of risk factors. The ICAPM implies that any state variables that characterize shifts in the investment and consumption opportunity sets are logical candidates. In practice, the choice of variables has been guided by both economic theory and extant empirical evidence. The APT assumption of a factor structure naturally lends itself to using multivariate statistical techniques, such as factor analysis and principal components, to construct factor-mimicking portfolios. The evidence generally indicates that the multi-factor models do a better job in explaining the cross-section of asset returns than the earlier CAPM literature, but that they are not capable of explaining all of the predictive power of variables such

as size [e.g., Roll and Ross (1980), Chen (1983), Chan, Chen and Hsieh (1985), Chen, Roll, and Ross (1986), Connor and Korajczyk (1988), and Lehmann and Modest (1988)].

Given the empirical success of size and book-to-market ratios in explaining asset returns, Fama and French (1993) propose a multi-factor model in which the additional factors are excess returns on portfolios that differ in these characteristics plus two factors related to interest rate and default risks. The model is:

$$E(r_i) = r_f + \beta_{i,m}[E(r_m) - r_f] + \beta_{i,SMB}[E(r_s) - E(r_B)] + \beta_{i,HML}[E(r_H) - E(r_L)] \\ + \beta_{i,TERM}[E(r_{LTG}) - r_f] + \beta_{i,DEF}[E(r_{CB}) - E(r_{LTG})]$$

where r_m and r_f are the returns on proxies for the market portfolio and riskless asset, as in the CAPM and ICAPM; r_s is the return on a portfolio of small-capitalization stocks, r_B is the return on a portfolio of large-capitalization stocks, r_H is the return on a portfolio of stocks with high book-to-market ratios, r_L is the return on a portfolio of stocks with low book-to-market ratios, r_{LTG} is the return on a long-term government bond, and r_{CB} is the return on a portfolio of long-term corporate bonds.

They find that the additional factors represent common sources of risk that are not subsumed in assets' relation with the market portfolio. They also find that the values of α_i for a set of portfolios formed by sorting assets by size and book-to-market ratios are much smaller when the additional factors are included.

Fama and French (1996) subject a variant of their proposed multi-factor model to a wide array of variables that have been able to predict deviations from the CAPM. Since the

bond market factors have little explanatory power for the cross-section of stock returns, a three-factor model is used:

$$E(r_i) = r_f + \beta_{i,m}[E(r_m) - r_f] + \beta_{i,SMB}[E(r_S) - E(r_B)] + \beta_{i,HML}[E(r_H) - E(r_L)].$$

In addition to firm size and book-to-market ratio, they test the model's ability to explain returns on portfolios formed on the basis of relative strength, earnings yield (earnings divided by price), and cash yield (cash flow divided by price). Fama and French find that the three-factor model does well against all of the predictive variables, except for the **Jegadeesh and Titman** relative strength variable.

The paper by **Brennan, Chordia, and Subrahmanyam** studies the predictive ability of the Fama and French three-factor model and the Arbitrage Pricing Theory where the factor-mimicking portfolios are estimated statistically by asymptotic principal components [Connor and Korajczyk (1986)]. In addition to predictive variables such as size, book-to-market equity, and relative strength proxies, they include proxies for liquidity (past trading volume) and tax effects (dividend yield) as possible non-systematic-risk-based predictors of differences in asset returns.

They find that both models fail to fully explain the predictive ability of size, book-to-market equity, relative strength, and past trading volume. The dividend yield variable cannot explain statistically significant deviations from the Fama-French model but can explain deviations from the version of the APT. For either model, the coefficient on past trading volume is negative, which is what one might expect if differential liquidity was priced and high past volume was a proxy for liquidity. This could also be consistent with the **Merton (1987)**

incomplete information model if an asset's past trading volume is correlated with the fraction of investors informed about that asset. Also, inclusion of past trading volume leads to a positive marginal size effect rather than the usual negative relation between size and returns. This might indicate that the predictive ability of size is related to liquidity.

While the CAPM, in theory, is very clear on the specification of the portfolio against which we measure systematic risk, the ICAPM provides a great degree of latitude in choosing the appropriate factors. This is both an advantage and a curse, particularly when the factors are chosen based on observed deviations from alternative models. The concern is that by choosing factor portfolios based on observed anomalies, we will be able to explain asset returns in sample but will not have any better out-of-sample performance. The Fama and French size and book-to-market factors were chosen, at least in part, on the basis of the evidence in **Fama and French (1992)** and the papers cited there. While it is difficult to completely rule out such a data mining effect, **Fama and French (1996)** cite various out-of-sample results (such as studies on different time periods or different national markets) that are consistent with the three-factor model.

Another approach to sorting out this issue is taken by **Daniel and Titman (1997)**. If a risk-based explanation is correct, then variation in systematic risk, as measured by $\beta_{i,SMB}$ and $\beta_{i,HML}$, rather than variation in the firm characteristics, size and book-to market, should explain variation in returns. That is, the risk-based model implies that a small firm with a low value of $\beta_{i,SMB}$ should have a low return. If returns are based solely on characteristics, rather than systematic risks, the small firm should have a high return regardless of its value of $\beta_{i,SMB}$. They study the differences in returns on portfolios that have the same size and book-to-market characteristics, but different values of the estimated value of $\beta_{i,HML}$. The risk-based model

implies that portfolios with high values of $\beta_{i,HML}$ should have higher returns than portfolios with low values of $\beta_{i,HML}$. Alternatively, if returns are based merely on the firm characteristics, rather than $\beta_{i,HML}$, portfolios matched on the characteristics should have the same return regardless of differences in their book-to-market betas. **Daniel and Titman** find that there is little variation in average returns for portfolios that have different values of $\beta_{i,HML}$ but similar size and book-to-market characteristics. They interpret the evidence as supporting the characteristics-based, rather than risk-based, model.

Davis, Fama, and French (1998) argue that the results of **Daniel and Titman** are specific to the time period they study. **Daniel and Titman** study the period from 1973 to 1993. Davis, Fama, and French (1998) combine book value data obtained from COMPUSTAT (post-1954) with data collected from Moody's Industrial Manuals in order to expand their sample to cover the period from 1925 to 1996. They perform the **Daniel and Titman** analysis on the period from July 1929 through June 1997 as well as various subperiods. They find that there is significant variation in average returns for portfolios that have different values of $\beta_{i,HML}$ but similar size and book-to-market characteristics, except for the 1973 to 1993 period.

The CAPM, ICAPM, and equilibrium APT models each imply that some combination of portfolios is on the mean-variance efficient portfolio frontier. The CAPM implies that the market portfolio is mean-variance efficient. The multi-factor pricing models imply that there is a combination of the factor mimicking, or market and state-variable, portfolios that is mean variance efficient. If the investment opportunity set is changing through time, then rational investors will choose portfolios based on the portfolio opportunity set, conditional on currently observed information. The empirical tests thus far have investigated unconditional mean-

variance efficiency rather than conditional mean-variance efficiency. While unconditional efficiency implies conditional efficiency, conditional efficiency does not imply unconditional efficiency [Hansen and Richard (1987)]. Thus, rejecting that a set of portfolios as unconditionally efficient does not imply that they are conditionally inefficient. In a portfolio management context, this distinction can be critical. By definition, active portfolios are based on dynamic trading strategies formed on the basis of (public and private) information about the conditional portfolio opportunity set. Portfolios that are adding value (positive values of α) relative to the conditional opportunity set may look inferior relative to the unconditional set. Similarly, portfolios that are adding no value relative to the conditional opportunity set (i.e., $\alpha = 0$) may look superior relative to the unconditional opportunity set formed from constant-weight combinations of assets in the economy. Therefore, we would like to evaluate asset pricing models and the performance of managed portfolios relative to the conditional opportunity set.

Ferson and Harvey estimate and test conditional versions of the ICAPM, allowing for time variation in the beta coefficients. In a conditional asset pricing model, returns on asset can be predictable to the extent that the systematic risk (betas) or the risk premia are time-varying. Return predictability that is not due to variation in betas or risk premia is evidence against the asset pricing model. **Ferson and Harvey** investigate the extent to which the observed time-series predictability of asset returns can be attributed to variation in betas and risk premia. They find that most of the predictability in asset returns is due to predictability in risk premia and systematic risk, with the risk premia being the more important of these two sources of predictability. Approximately 10% of the return predictability, for their set of assets, is due to non-risk factors. Conditional versions of the ICAPM and the APT are also tested in Shanken

(1990) and Ferson and Korajczyk (1995). While implementing conditional asset pricing models is significantly more complicated than implementing unconditional models, accommodating conditioning information will be important for accurately measuring the performance of actively managed portfolios.

Market Imperfections and Asset Pricing

The models we have considered so far, with the exception of **Merton (1987)**, are based on a paradigm of frictionless markets. We have ignored trading costs, taxation, and asymmetric information. One could motivate the incomplete information model of **Merton (1987)** by the existence of such frictions. That model predicts that such incomplete information leads to non-zero values of assets' alphas relative to models derived assuming no frictions. For example, firms with fewer informed investors will have a positive α relative to the CAPM predictions. If small-capitalization firms are more likely to have relatively fewer informed investors, the model provides one explanation for the small-firm anomaly where small firms have higher returns than large firms after adjusting for systematic risk. In this section we look at two frictions that may cause investors to price assets differently than predicted by the frictionless market models. Those frictions are transactions costs and non-tradability of certain assets.

Amihud and Mendelson derive a model in which the cost of trading a given asset is the spread between its bid and ask prices. Investors sort across assets on the basis of the length of their expected holding period. Investors with shorter (longer) expected holding periods will hold assets with a lower (higher) percentage bid-ask spread. In equilibrium, assets with a higher

spread will sell for a lower price, leading to a higher expected rate of return before transactions costs. The higher rate of return compensates for the higher expected transactions costs.

Empirically, they find that average returns on stocks increase with the percentage spread after controlling for differences in $\beta_{i,m}$. Given that trading volume and the percentage spread are likely to be correlated, this result is related to the relation between returns and past trading volume found by **Brennan, Chordia, and Subrahmanyam**. **Amihud and Mendelson** also test whether the size anomaly persists when one accounts for the bid-ask spread. They find that the bid-ask spread's ability to predict cross-sectional differences in asset returns remains while firm size is statistically insignificant.

The effects of liquidity are investigated further in the next paper in this section, **Brennan and Subrahmanyam (1996)**. They measure illiquidity by looking at the impact of intra-day trading volume on prices. Fixed and variable components of the price impact are estimated. They find that there is a significant positive relation between α_i from the three-factor Fama/French model and their measures of the fixed and variable components of the price impact. They also find a negative relation between α_i and the proportional bid-ask spread. This latter finding is at odds with the results from **Amihud and Mendelson**.

The extant evidence indicates that liquidity seems to be reflected in differences in expected returns across traded securities. Whether this reflects true pricing of liquidity, correlation between the liquidity proxies and omitted sources of risk, or sample specific effects, will require further research.

In the frictionless-world asset pricing models it is assumed that all assets are tradable. Human capital, the value of one's earnings stream, is an asset that is not tradeable. For many

investors, the value of their human capital is large relative to the value invested in tradable assets. The risk of that human capital is likely to give rise to hedging demands on the part of investors. To the extent that asset prices are correlated with shocks to the value of human capital, the hedging demands should lead to prices, or expected rates of return, different from those predicted by the models that ignore the non-tradability of human capital. Mayers (1973) derives an asset pricing model that incorporates non-tradable human capital. The equilibrium expected rates of return look much like a two-factor ICAPM where one factor is the market portfolio and the other factor is the aggregate payoff to human capital.

Jagannathan and Wang estimate a conditional version of the CAPM that incorporates the risk associated with human capital. One possible interpretation of including human capital in the analysis is that one is obtaining a better proxy for the market portfolio in a standard CAPM model. Alternatively, inclusion of a separate human capital factor is consistent with the separate hedging demands caused by the non-tradability of human capital, as in Mayers (1973). Much like **Fama and French (1992)**, **Jagannathan and Wang** find that there is no relation between actual and predicted returns from the CAPM using a standard stock portfolio as the proxy for the market (see their Figure 1) and returns exhibit a very pronounced size effect. However, when they add a factor for human capital, there is a very strong relation between actual and predicted returns (see their Figure 3) and firm size has no significant explanatory power for predicting α_i . Similar results are found for the determinants of expected returns on common stocks in Japan [Jagannathan, Kubota, and Takehara (1998)].

The apparent success of a human capital factor in explaining returns suggests that incorporating the risk of the payoff on human capital should be an important component in asset

pricing models. The human capital augmented pricing models imply that, in addition to traditional diversification, investors avoid assets that are correlated with shocks to their labor income. However, there is evidence that investors are not as well diversified as the models predict and much of this lack of diversification is related to entrepreneurial income. Heaton and Lucas (1997) find that a variant of the **Jagannathan and Wang** model that incorporates entrepreneurial income risk outperforms a variant that focuses on labor income risk.

Portfolio Performance Evaluation

Asset pricing models provide a benchmark against which managed portfolios can be compared. It is obvious that an S&P 500 index fund's return cannot be compared directly to the return on a money market fund. The stock fund is considerably riskier and we know that, in the long run, stocks have significantly outperformed money market instruments. The question is how much higher a performance hurdle should we set for the index fund versus the money market fund. The answer provided by the CAPM, ICAPM, and APT is that funds with higher systematic risk, that is higher values of beta relative to the market, the ICAPM state variables, or the APT pervasive factors, should have higher returns. Therefore, we should compare a managed portfolio's return to the return on a passive portfolio that has the same level of systematic risk. This is exactly what our pricing deviation parameter, α_i , does. It is the difference between the return on the managed portfolio and the return on a passive portfolio with the same level of systematic risk.

Alpha, α_i , was used in this context in Jensen (1968) and has come to be known as the Jensen measure of portfolio performance. A manager who has superior ability in identifying

assets with positive and negative values of α_i should be able to form a portfolio with a positive alpha by buying the positive alpha assets and selling the negative alpha assets. On the other hand, a manager trading with insufficient ability may not cover the portfolio's transactions costs, resulting in a negative value of alpha after transactions costs. It is important to note that we do not observe the true value of alpha, but only noisy estimates of alpha, $\hat{\alpha}_i$, based on historical returns. Over short intervals good managers could have negative estimates of alpha just by chance and inferior managers could have positive values of $\hat{\alpha}_i$ just by chance. Therefore, the performance literature has focused on separating skill from luck by looking for statistically significant performance.

For a managed portfolio with a given level of α_i , it is possible to create portfolios with higher or lower alphas by leveraging and delevering the portfolio. That is, by borrowing or lending at the riskless rate of interest. Thus, the alphas of this set of portfolios are determined by (i) the quality of the manager's information and (ii) how aggressively the manager acts on this information (i.e., the leverage). Treynor (1965) suggests an alternative measure in a CAPM context: the ratio of α_i to $\beta_{i,m}$ (plus a constant). The effects of leverage influence α_i and $\beta_{i,m}$ by the same proportion, so that the ratio is independent of leverage, giving a measure that is based on the quality of the manager's information and not on the aggressiveness with which that information is implemented. This measure is known as the Treynor measure of performance.

Is it optimal for investors to choose to invest all of their wealth in the actively managed portfolio with the highest alpha? The answer is no. The intuition behind the asset pricing models is that only systematic risk matters because investors are optimally choosing portfolios that eliminate unsystematic risk (absent access to superior information). The portfolio with the

largest value of alpha may have a large amount of unsystematic risk which cannot be diversified away if one places all of one's wealth in that portfolio. An investor's optimal portfolio will be based on the correct tradeoff between obtaining a higher alpha and taking on more unsystematic risk. If the investor has mean-variance preferences, the optimal portfolio is one that maximizes the ratio of expected return (in excess of the riskless asset) to the standard deviation of the portfolio. This metric is known as the Sharpe measure of portfolio performance [Sharpe (1966)] and is an appropriate measure for the investors' total portfolio rather than the individual assets or funds that make up that portfolio. Treynor and Black [1973] show how one optimally uses the information conveyed by alpha to create a portfolio with maximal Sharpe measure.

A portfolio manager's true, forward-looking value of α_i is not observable. For most portfolios with a track record we can observe an estimate of historical performance, $\hat{\alpha}_i$. Much work has been done on the estimation of alpha and the relation between past performance (measured by $\hat{\alpha}_i$) and future performance. The evidence is that mutual funds tend to have negative values of alpha on average. Statistically significant values of $\hat{\alpha}_i$ are rare. However, in most cases the size of the estimation error in $\hat{\alpha}_i$ implies that abnormal performance needs to be economically very large in order to be statistically significant.²

There is some controversy about whether historical performance has *any* predictive power for future performance. Many earlier studies found that there is no relation between performance in one period and performance in subsequent periods. More recent studies [e.g.,

²Estimates of the alphas, relative to the unconditional CAPM, for a sample of mutual funds are given in Jensen (1968). Lehmann and Modest (1987) and Connor and Korajczyk (1991) compare CAPM-based and APT-based measures of alpha.

Grinblatt and Titman (1992); Brown and Goetzmann (1993); Elton, Gruber, Das, and Hlavka (1993); and Goetzmann and Ibbotson (1994), Christopherson, Ferson, and Glassman (1998)] indicate that, while there is considerable reversion to the mean, good performance tends to be followed by good performance and bad performance tends to be followed by bad performance. The latter effect tends to be stronger. That is, persistence in performance is stronger for under-performers than for good performers.

Mirroring the literature on testing asset pricing models, the portfolio performance evaluation literature was initially applied in an unconditional setting. However, the essence of many active portfolio strategies is that the composition of the portfolio is altered in response to information about the portfolio opportunity set. “Stock picking” or “security selection” portfolio managers are choosing individual assets on the basis of their predictions of those assets’ performance, α_i . “Market timing” or “sector rotation” portfolio managers change the level of systematic risks (the betas) on the basis of their predictions of the risk premia for those factors. Additionally, portfolios that include derivative securities or use dynamic trading strategies designed to replicate derivative securities will lead to portfolios whose risk characteristics can vary significantly through time. Portfolio insurance, the strategy of holding assets and put options on those assets - or the dynamic equivalent of that strategy, is an example. Even in situations where the portfolio opportunity set, conditional on public information, is constant through time, it turns out that the performance of certain types of active portfolio management or dynamic trading strategies cannot be measured accurately with the standard unconditional asset pricing models.

If we observe the manager's choice of portfolio composition each period (and the observation period is at least as fine as the portfolio manager's investment decision periods) we can decompose historical performance into its various components, along the lines of Fama (1972). However, it is often the case that timely data on portfolio composition are not available. This forces us to make inferences based on the history of returns on the managed portfolio.

Jensen (1972) and Dybvig and Ross (1985) show that while portfolio managers' access to security specific information can be accommodated by unconditional models, the existence of market timing or sector rotation ability can lead to incorrect inferences about the α of the portfolio. That is, while the manager is generating a positive value of α relative to the conditional CAPM relation, unconditionally, α could be either positive or negative. Additionally, the unconditional Sharpe measure could indicate that this superior manager is below the mean/variance efficient frontier. This problem is caused by the fact that shifts in systematic risk, based on the manager's private information, may be confused with higher unconditional risk by performance measures that look at the unconditional distribution of returns on the portfolio. Also, even if individual asset returns are normally distributed, both conditionally and unconditionally, the managed portfolio's unconditional return is not normally distributed when the manager has market timing ability. There is positive skewness in the portfolio return that is valued by investors but which is not reflected in the mean-variance metrics.

The inclusion of options in the portfolio causes similar non-normality of portfolio returns that is not well-accommodated by unconditional performance measure. For example, Dybvig and Ingersoll (1982) show that in an economy in which the primary assets are priced by the

unconditional CAPM, options on those primary assets will not be properly priced by the CAPM (i.e., they will exhibit non-zero values of α even though they are fairly priced).

Figure 1 of the paper by **Grinblatt and Titman** gives a clear illustration of the potential problem caused by timing ability. They decompose Jensen's measure of performance, α_i , into three components: security selection ability, market timing ability, and bias in measured systematic risk. The last component is the culprit causing misestimation of performance when the portfolio manager is a market timer.

Grinblatt and Titman propose the Positive Period Weighting Measure (PPWM) in which performance is measured by a weighted average of the portfolio's returns. The weights are constructed so that they are positive each period and the weighted average of the market portfolio is zero. They show that the PPWM will assign zero performance to uninformed investors, positive performance to managers with stock picking ability, and positive performance to managers with timing ability (as long as their timing strategy is one where systematic risk is increasing in their forecast of the market excess return). Thus, the measure eliminates the term in the Jensen measure due to biases in systematic risk caused by timing ability.

An alternative approach to measuring performance that accounts for timing ability is to specify the relation between the portfolio managers' market information and their choice of systematic risk. This allows an augmented model that separates the measurement of timing and selection ability. Examples of this are Treynor and Mazuy (1966), Jensen (1972), Henriksson and Merton (1981), and Pfleiderer and Bhattacharya (1983). Empirically, most studies find that the average managed portfolio studied exhibits negative measured timing ability. Cross

sectionally, there is negative correlation between measured market timing ability and measured stock picking ability [e.g., Henriksson (1984) and Shukla and Trzcinka (1994)].

As noted above, timing ability causes problems for unconditional performance measures in Dybvig and Ross (1985) because of the non-normality created in the distribution of asset returns. The inclusion of options in the portfolio causes similar non-normality of portfolio returns that is not well-accommodated by unconditional performance measures. In the absence of timing ability one could create false measured timing by purchasing options on the market portfolio (or other option-like securities) or dynamically trading assets in the portfolio to replicate the payoff on such options. True timing ability is similar to receiving a free option on the market portfolio [Merton (1981)]. Creating false timing through purchasing the option, on the other hand, leads to a negative measure of the security-specific component of performance, which offsets the positive timing performance. This is consistent with the empirical evidence that finds negative correlation between measured security selection and timing performance.

Jagannathan and Korajczyk show how to combine the selection and timing measures for a Henriksson and Merton type of market timer. Whether a manager is following a Henriksson and Merton timing strategy or creating fake timing by buying options like those created by a Henriksson and Merton timer, the combined measure will give an appropriate measure of total performance. In particular, a manager creating fake timing ability by buying options will have negative security selection performance that exactly offsets the timing performance. **Jagannathan and Korajczyk** show that this result is sensitive to deviations from the exact Henriksson/Merton timing strategy. That is, something as simple as creating false timing by buying options with a different exercise price from that assumed in Henriksson and

Merton (1981) can lead to non-zero performance for this manager, even though no true ability exists in this strategy. Thus, getting the manager's strategy right is important in these types of measures. Glosten and Jagannathan (1994) propose using a series of options to approximate the manager's portfolio strategy.

We have seen that non-normality in portfolio returns, whether due to holding option-like securities, replicating option through dynamic trading, or through timing ability, can lead to misclassification of a portfolio manager's ability. A number of alternative approaches to non-normality have been proposed. Markowitz (1959) discusses semi-variance rather than variance as a risk measure. The variance and standard deviation treat upside and downside uncertainty equally. This is not a problem when dealing with symmetric distributions since the upside and downside uncertainties are the same. With the non-symmetric returns induced by options and market timing strategies, the downside risk may differ from the upside potential. Semi-variance is a weighted average of squared deviations below the mean. That is, deviations above the mean are treated as being equal to zero. Related risk measures include Downside Deviation, Lower Partial Moments and related risk-return ratios [Balzer (1994) and Sortino and Lee (1994)]. Harlow and Rao (1989) develop an asset pricing model based on lower partial moments, rather than variance as measures of overall portfolio risk. Their empirical specification is almost identical to that of Henriksson and Merton (1981) with the exception that (i) the exercise price of the option created by the portfolio manager's timing ability (or the exercise price of the options purchased to create false timing) is estimated (Merton and Henriksson assume that the exercise price is higher than the current market index by a factor equal to the riskless interest rate), and

(ii) the intercept is allowed to vary as a function of whether the market is “up” or “down” (i.e. whether the aforementioned options are in- or out-of-the-money).

Leland shows how a dynamic trading strategy implies that the market portfolio is mean-variance inefficient relative to a portfolio opportunity set that includes the payoffs of dynamic trading strategies. This also implies that assets’ CAPM pricing deviations, α_i , will not be zero. **Leland** goes on to develop a performance measure [based on the results of Rubinstein (1976)] that can accommodate portfolios that have arbitrary return distributions, including those created by holding derivative securities within the portfolio. The modified alpha is calculated as the pricing deviation, as in the CAPM, except that the definition of systematic risk for a portfolio, say p, changes from:

$$\beta_{p,m} = \frac{\text{cov}(r_p, r_m)}{\text{var}(r_m)}$$

to:

$$B_{p,m} = \frac{\text{cov}(r_p, -(1+r_m)^{-b})}{\text{cov}(r_m, -(1+r_m)^{-b})}$$

where the parameter, b, is the Sharpe ratio for the market portfolio. **Leland** shows that the redefined alpha is zero for option-based strategies that yield non-zero values of alpha using the CAPM beta. Thus, the performance measure can lead to substantially improved measures of performance for portfolios containing derivative securities positions. The derivation of the new measure requires that the return on the market portfolio be independently and identically distributed (i.i.d.) through time and that the redefined measure of systematic risk is constant.

Thus, it is very much in the spirit of the unconditional CAPM. There is evidence that the returns on standard proxies for the market portfolio are not i.i.d. and there is no reason to expect that any given portfolio has a modified risk parameter, $B_{p,m}$, that is constant through time. Whether these deviations from the model's assumptions lead to large or small errors is an empirical question that deserves further attention.

An alternative approach, taken in **Ferson and Schadt**, is to explicitly condition systematic risk estimates and portfolio performance on available information. There is evidence that public information can be used to predict changes in the portfolio opportunity set (e.g., some studies find that the market risk premium varies with the business cycle). In this case, unconditional performance measures could attribute superior performance to portfolio strategies that could be replicated by anyone since they are based solely on public information. **Ferson and Schadt** use conditional versions of the CAPM and ICAPM that explicitly model changes in systematic risk as a function of public information. They also estimate conditional versions of the Treynor and Mazuy (1966) and Henriksson and Merton (1981) timing measures for the CAPM.

They estimate unconditional and conditional performance measures for a sample of mutual funds. For both the CAPM and ICAPM, unconditional Jensen measures, α_i , are negative on average while the average conditional Jensen measure is zero. Similarly, the timing measures tend to be negative using the unconditional model and positive using the conditional model. The difference in the unconditional and conditional results indicates that the relation between managers' choice of systematic risk and future returns on the market portfolio, *due to public information*, is negative on average. This odd result could be due to an explicit decision to shift toward low beta stocks when public information indicates a high risk premium or due to risk

changes in the underlying assets that are not offset by an explicit change in portfolio composition. **Ferson and Schadt** find that the same anomalous pattern occurs for a simulated passive buy-and-hold policy. Since such a passive policy is not driven by a manager choosing to change the portfolio composition, the changes in portfolio risk are due to changes in the risk of the underlying assets. This might point toward the latter effect driving the results for the mutual funds. **Ferson and Schadt** also point to some evidence that suggests that the anomalous movement in betas is due to the timing of new inflows of capital to the funds.

The final paper in this volume, by **Brown, Goetzmann, Ibbotson, and Ross**, takes a new look at the literature about the predictive power of past portfolio performance for future portfolio performance discussed earlier. Performance persistence is measured by looking at the correlation, across funds, between prior period performance and subsequent period performance. The issue here is that this correlation cannot be computed for funds that existed in the prior period, but went out of business, so that data do not exist in the subsequent period. Thus, the persistence tests have an embedded survivorship bias. **Brown, Goetzmann, Ibbotson, and Ross** show that this survivorship bias leads to an upward bias in estimates of predictability. Since there is no simple adjustment available to correct measured persistence for this bias, based on observable data, the results mainly serve to caution us about over-interpreting empirical results on performance persistence.

The task of evaluating of actively managed portfolios presents us with many challenges. This is particularly true when the risk of the portfolio is changing (whether through market timing

and sector rotation strategies or through holding derivative securities). The papers here provide us with a set of tools that help us distinguish between skill, risk taking, and luck.

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